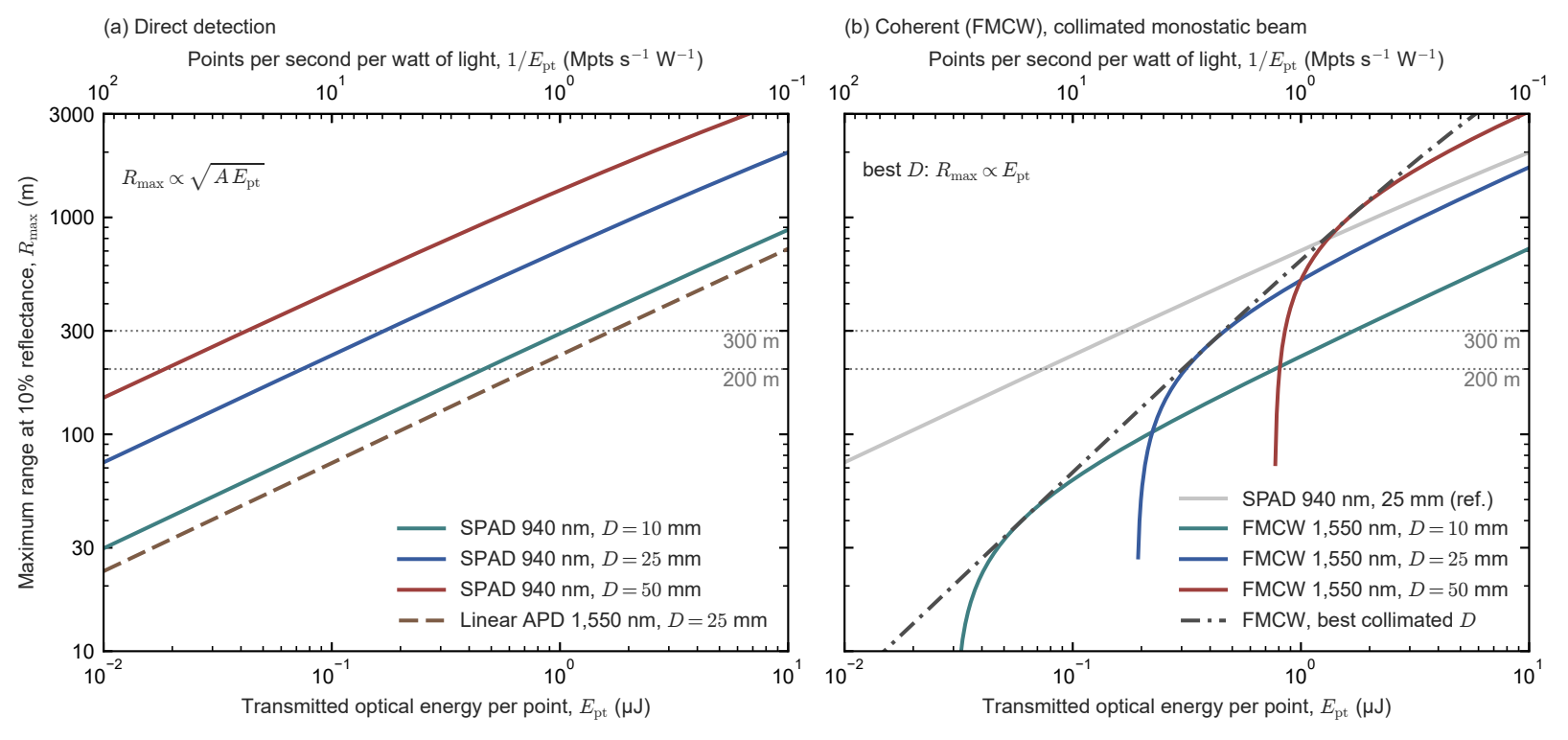
(a) Direct detection
Points per second per watt of light, $1/E_{\mathrm{pt}}$ (Mpts s^{-1} W^{-1})
10^{2}
10^{1}
10^{0}
10^{-1}
3000
1000
300
100
30
10
Maximum range at 10% reflectance, R_{max} (m)
$R_{\mathrm{max}} \propto \sqrt{A\,E_{\mathrm{pt}}}$
300 m
200 m
SPAD 940 nm, $D = 10$ mm
SPAD 940 nm, $D = 25$ mm
SPAD 940 nm, $D = 50$ mm
Linear APD 1,550 nm, $D = 25$ mm
10^{-2}
10^{-1}
10^{0}
10^{1}
Transmitted optical energy per point, E_{pt} (μJ)
(b) Coherent (FMCW), collimated monostatic beam
Points per second per watt of light, $1/E_{\mathrm{pt}}$ (Mpts s^{-1} W^{-1})
10^{2}
10^{1}
10^{0}
10^{-1}
best D: $R_{\mathrm{max}} \propto E_{\mathrm{pt}}$
300 m
200 m
SPAD 940 nm, 25 mm (ref.)
FMCW 1,550 nm, $D = 10$ mm
FMCW 1,550 nm, $D = 25$ mm
FMCW 1,550 nm, $D = 50$ mm
FMCW, best collimated D
10^{-2}
10^{-1}
10^{0}
10^{1}
Transmitted optical energy per point, E_{pt} (μJ)